SFIT Stochastic-Braid Substrate: Part III

Finite-carrier record formation and the conditions for Born-weighted outcomes: a structural analysis

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Abstract

We separate three objects that are easily conflated in attempts to obtain quantum measurement statistics from a classical substrate: the configuration of the substrate, the population of a measurement channel, and the probability that a record names that channel. Three questions follow. (i) Can a shared, finite carrier enforce exclusive records? In a coupled-mode model with explicit local detectors, a local back-reaction (a detector that fires increases its own coupling) lowers the gated coincidence parameter from $\alpha = 1.00$ to $\alpha = 0.57$–0.60 at the strongest back-reaction tested, but only when detector thresholds are comparable to the carrier energy; at thresholds one twentieth of the carrier energy, α stays at 0.96–1.00. Strong exclusion ($\alpha < 0.1$) was not reached in any of 136 configurations. (ii) Under what conditions does record selection reproduce Born weights? We examine two routes. Memoryless (exponential) thresholds give Born frequencies exactly, by a common-time-change argument; gamma thresholds of shape 2 depart by a total-variation distance of 0.08–0.15. A bounded martingale for the channel populations that is absorbed at a vertex gives Born weights for any functional form of the exchange law; we prove a first-passage bound and a converse (given absorption, zero conditional drift is necessary), and show with a biased exchange kernel that the recorded frequencies follow the Kimura fixation formula, so that zero drift must hold to order $1/M$ in the number of exchange units. Noise acting on the shared carrier cannot substitute for detector-local randomness: with fixed thresholds the largest channel fires first in every run, at every reversal rate. (iii) Does Part I supply these conditions? No. The helicity representation of Part I acts by permutation matrices, and we prove that no permutation action yields zero conditional drift for any filtration unless it is the identity; measured on the Part I crossing process, the drift is nonzero at every test state, and populations are rearranged but never concentrated. Sector populations are exactly invariant, which identifies braiding as neutral with respect to record selection. No derivation of the Born rule or of quantum superposition from the substrate is claimed.

1 Scope

Part I [1] showed that persistent stochastic defects in the plane generate braid-group structure, and Part II [2] identified conditions under which a classical absorber race produces Born-rule record frequencies. Part II left two questions open: whether records can be made mutually exclusive by local physics, and whether the substrate supplies the statistical conditions that the race requires. This paper addresses both, and adds a third route to Born weights that Part II did not consider.

We distinguish:

- the *substrate configuration*: positions, velocities and helicities of the defects of Part I;

- the *channel population* $x_k \geq 0$, $\sum_k x_k = 1$, of a measurement channel k; when an amplitude ψ_k is given, $x_k = |\psi_k|^2$;
- the *record probability* $P(k)$: the probability that a physical record names channel k.

The Born rule is the statement $P(k) = |\psi_k|^2$. It relates the third object to the second; it says nothing by itself about how the second arises from the first. Table 1 summarises what this paper finds. Throughout, each result is tagged [DERIVED] (follows from stated equations), [ASSUMED] (an assumption of the model, not a consequence of Part I) or [TESTED] (established by simulation with controls).

Question	Status
Can a shared finite carrier enforce exclusive records?	Partial suppression only ($\alpha = 0.57$–0.60 at best), and only for thresholds comparable to the carrier energy. Strong exclusion not achieved.
Under what conditions does record selection reproduce Born weights?	Two sufficient routes identified, each with a precise fragility: exactly memoryless thresholds, or zero conditional drift (to order $1/M$) with absorption. Neither is shown to be the only route.
Does Part I supply those conditions?	No. The Part I helicity representation is a permutation action and cannot supply a population martingale for any filtration.

Table 1: The three questions and their status.

2 Definitions

Records and frequencies. A run produces zero or more detector firings. The *first-record frequency* is $P(k) = \Pr(\text{detector } k \text{ fires first} \mid \text{at least one fires})$. The fraction of runs with no firing is reported separately. The Born test is an equivalence test: the upper end of a 99% bootstrap interval for the total-variation distance $\mathrm{TV}(P, |\psi|^2)$ must lie below $\varepsilon = 0.02$.

Gated coincidence. Each run is one gate. With P_i the probability that detector i fires during the run and P_{ij} the probability that both i and j fire,

$$\alpha = \frac{\sum_{i \neq j} P_{ij}}{\sum_{i \neq j} P_i P_j}. \tag{1}$$

Independent detectors give $\alpha = 1$; a classical field split among independent detectors gives $\alpha \geq 1$; single-photon experiments give $\alpha \approx 0$ [3]. We call $\alpha_{\mathrm{hi}} < 1$ (upper 99% bound) *partial* exclusion and $\alpha_{\mathrm{hi}} < 0.1$ *strong* exclusion. Equation (1) is undefined when fewer than two detectors can fire; such configurations are reported as undefined, not assigned a value.

3 Question 1: exclusive records from a shared finite carrier

3.1 Model

A single carrier mode with complex amplitude a holds energy $|a|^2$, initially $E_0 = 1$, and an internal state ψ of unit norm [ASSUMED]. Channel k has a local damped detector oscillator b_k. In the

rotating frame,

$$\dot{a} = -i \sum_k \bar{G}_k b_k, \qquad \dot{b}_k = -iG_k a - \tfrac{\gamma}{2} b_k, \qquad \dot{W}_k = \gamma |b_k|^2, \tag{2}$$

with $G_k = g\,\kappa_k\,\psi_k$. The coupling is linear in ψ_k [ASSUMED]. Energy is conserved, $|a|^2 + \sum_k |b_k|^2 + \sum_k W_k = E_0$ [DERIVED]; the numerical error was below 7.4×10^{-7} in every run. Detector k fires when its absorbed energy W_k crosses a threshold H_k [ASSUMED]. On firing, and only then, detector k raises its *own* coupling, $\kappa_k : 1 \to \kappa \geq 1$ [ASSUMED]. No rule ends a run, blocks another detector or selects a winner. Substrate noise enters as Poisson reversals of the carrier phase, $a \to -a$, at rate ν, standing for the velocity reversals of Part I [ASSUMED].

Parameters: $g = 1$, $\gamma = 4$, $\kappa \in \{1, 2, 4, 8\}$, $\nu \in \{0, 1, 5\}$, threshold scale $\theta \in \{0.05, 0.3\}$; thresholds fixed at θ, exponential with mean θ, or gamma of shape 2 and mean θ. Five states were used: $|\psi|^2 = (0.2, 0.8)$, $(0.5, 0.3, 0.2)$, $(0.7, 0.2, 0.1)$, $(0.4, 0.3, 0.2, 0.1)$, and $(0.5, 0.3, 0.2)$ with complex phases. Integration is RK4 with $\Delta t = 0.01$; first crossings that fall in the same step are re-integrated with 50 sub-steps, leaving a residual ambiguity of at most 5×10^{-4} of records (budget 10^{-3}). Each configuration has 20 000 runs.

3.2 What the model derives

Proposition 1 (Common time change)**.** *Before the first firing, $b_k(t) = G_k\,\beta(t)$ for a single function β, so $W_k(t) = |\psi_k|^2\,U(t)$ with U common to all channels, whatever the reversal history.*

Proof. With $b_k(0) = 0$, the equations for b_k are invariant under $b_k \mapsto G_k\beta$: substituting gives $\dot{\beta} = -ia - \frac{\gamma}{2}\beta$, independent of k, and $\dot{a} = -i\big(\sum_k |G_k|^2\big)\beta$. Reversals multiply a by -1 and do not break the form. Then $\dot{W}_k = \gamma g^2 |\psi_k|^2 |\beta|^2$. [DERIVED] □

The quadratic dependence of absorbed energy on ψ_k is therefore a consequence of linear coupling, not an input; the measured exponent $d \ln W_k / d \ln |\psi_k|$ was 2.00000 in every configuration [TESTED]. Proposition 1 has two consequences.

Corollary 1 (Common-mode noise cannot randomise the order)**.** *With fixed thresholds $H_k = \theta$, the channel with the largest $|\psi_k|^2$ fires first in every run, for any reversal history.* [DERIVED]

Observed: the largest channel fired first in 100% of runs in all 48 fixed-threshold configurations, for $\nu = 0, 1, 5$ and both θ [TESTED]. Randomness that makes record selection depend on $|\psi_k|^2$ rather than on its rank must therefore be local to the detectors; noise acting on the shared carrier is common-mode and cannot provide it.

Corollary 2 (Exponential thresholds give Born)**.** *With independent exponential thresholds of common mean, $P(k) = |\psi_k|^2$ exactly.* [DERIVED]

Proof. Detector k fires first iff $H_k / |\psi_k|^2$ is the smallest of the ratios $H_j / |\psi_j|^2$, which are independent exponentials with rates proportional to $|\psi_j|^2$. The monotone time change U does not alter the order, and truncation at finite total energy does not alter the conditional probability. □

Observed: for exponential thresholds, the Born test passed in all 48 configurations (largest upper TV bound 0.0172) [TESTED].

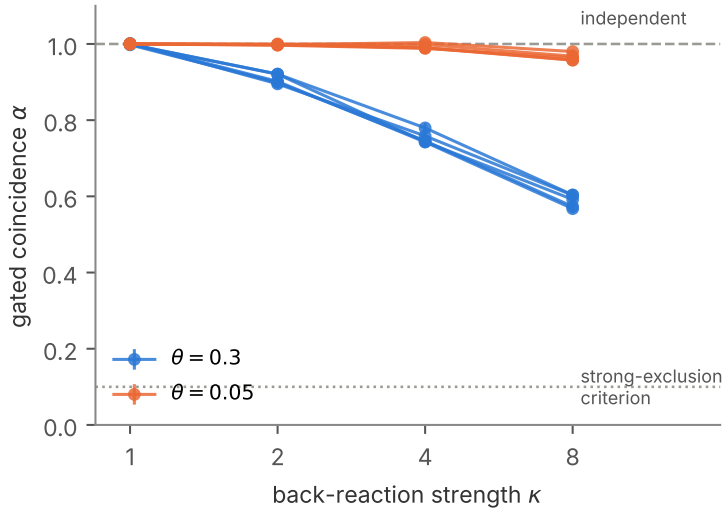


Figure 1: Gated coincidence α against back-reaction strength, exponential thresholds, $\nu = 1$. Each line is one of the five states; bars are 99% bootstrap intervals (mostly smaller than the markers). Blue: thresholds comparable to the carrier energy ($\theta = 0.3$). Orange: small thresholds ($\theta = 0.05$). Dashed: independent detectors. Dotted: strong-exclusion criterion.

3.3 Exclusion

Without back-reaction ($\kappa = 1$), the finite shared carrier alone gives $\alpha = 0.998$–1.001: a fixed energy split among detectors with independent thresholds produces independent firings [TESTED]. With $\theta = 0.3$, back-reaction lowers α monotonically: 0.895–0.921 at $\kappa = 2$, 0.743–0.780 at $\kappa = 4$, 0.568–0.604 at $\kappa = 8$, while the first-record frequencies remain Born (Fig. 1) [TESTED]. The mechanism is local: a detector that has fired drains the shared carrier faster, so the others receive less. Reversals strengthen it (for $(0.5, 0.3, 0.2)$, $\kappa = 4$: $\alpha = 0.789$, 0.743, 0.644 at $\nu = 0, 1, 5$).

Two limitations are part of the result.

Threshold scale. With $\theta = 0.05$, detectors fire 1.4–3.9 times per run and α stays at 0.957–1.004. The prediction, fixed before the run, that α decreases monotonically with κ at both threshold scales was *not confirmed*: for $(0.2, 0.8)$, $\alpha = 1.000$ at both $\kappa = 1$ and 2; for $(0.7, 0.2, 0.1)$, $\alpha = 1.004$ at $\kappa = 4$, above its value at $\kappa = 2$. When thresholds are small, every detector fires before back-reaction can starve it. Exclusion in this model requires thresholds comparable to the carrier energy.

Strength. Strong exclusion ($\alpha_{\text{hi}} < 0.1$) was reached in none of the 136 configurations; the lowest value was $\alpha = 0.537$ $[0.523, 0.552]$. The model produces partial suppression of coincidences, not the near-perfect exclusion of single-quantum experiments.

Controls. All six passed (20 000 runs each): synthetic independent detectors, $\alpha = 0.999$ $[0.984, 1.011]$; fixed thresholds without reversals, largest first in 100%; ψ-independent coupling, uniform frequencies (TV upper bound 0.013); classical limit $\kappa = 1$, $\alpha = 1.000$ $[0.996, 1.006]$; an imposed $|\psi|^2$ draw with a single firing, Born and $\alpha = 0$; ψ shuffled between channels and compared with the original target, mismatch detected (TV lower bound 0.289).

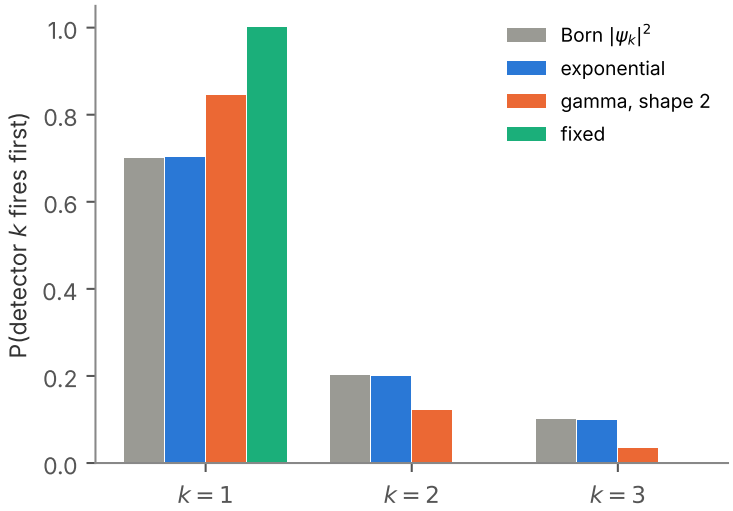


Figure 2: First-record frequencies for $|\psi|^2 = (0.7, 0.2, 0.1)$, $\theta = 0.3$, $\kappa = 4$, $\nu = 1$, under three threshold laws. Only exponential thresholds reproduce the Born values (grey).

4 Question 2: conditions for Born weights

4.1 Route 1: memoryless thresholds

Corollary 2 requires exponential thresholds. This is the content of Yellott's theorem [4] in the present setting: within races with independent, identically distributed thresholds, choice probabilities proportional to the rates for three or more alternatives (Luce's choice axiom [5]) require exponential thresholds. In the coupled-mode model, gamma thresholds of shape 2 gave TV distances of 0.082–0.148, always biased toward the largest channel; fixed thresholds gave the largest channel with certainty (Fig. 2) [TESTED]. A diagnostic separates the laws without reference to the target: the cumulative hazard of the first firing against absorbed energy is linear for a memoryless law. The ratio of its slopes on the first and second halves of the energy range was 0.97–1.04 for exponential thresholds and 0.50–0.56 for gamma thresholds [TESTED].

The route is fragile in its *shape*: any departure from an exactly memoryless law biases the records toward the largest channel. That the substrate produces exactly memoryless detector thresholds is [ASSUMED] and not shown here.

4.2 Route 2: martingale populations with absorption

Let $x(t) \in \Delta^{n-1}$ be the channel populations, adapted to a filtration $(\mathcal{F}_t)$.

Theorem 1 (Born weights from a martingale)**.** *If each x_k is a martingale and $x(t) \to e_K$ almost surely for a random index K, then $\Pr(K = k \mid \mathcal{F}_t) = x_k(t)$ for all t; in particular $\Pr(K = k) = x_k(0)$.*

Proof. x_k is bounded in $[0, 1]$, so it converges almost surely and in L^1, and $E[x_k(\infty) \mid \mathcal{F}_t] = x_k(t)$ [6]. Absorption gives $x_k(\infty) = \mathbf{1}_{\{K=k\}}$. [DERIVED] □

Proposition 2 (Converse)**.** *If $x(t) \to e_K$ almost surely and $\Pr(K = k \mid \mathcal{F}_t) = x_k(t)$ for all t, then x_k is a martingale.*

Proof. $x_k(t) = E[\mathbf{1}_{\{K=k\}} \mid \mathcal{F}_t]$ is a Doob martingale. [DERIVED] □

Given absorption, zero conditional drift is therefore not one sufficient condition among several: it is equivalent to the records being Born-weighted conditionally at every time. A measured nonzero drift excludes this route.

Absorption at a vertex is a limit and may not occur in finite time. A record defined by first passage is the operational version.

Proposition 3 (First-passage bound)**.** *Let $h > 1/2$, let τ be the first time some $x_k \geq h$, assume $\tau < \infty$ almost surely, and let K be that channel. If each x_k is a martingale, then $|\Pr(K = k) - x_k(0)| \leq (1 - h)/h$.*

Proof. At most one component can exceed $h > 1/2$, so K is well defined. By optional stopping for bounded martingales, $x_k(0) = E[x_k(\tau)]$. On $\{K = k\}$, $x_k(\tau) \in [h, 1]$; otherwise $x_k(\tau) \leq 1 - h$. Hence $h\,p \leq x_k(0) \leq p + (1 - h)(1 - p)$ with $p = \Pr(K = k)$, which gives $p \geq x_k(0) - (1 - h)$ and $p \leq x_k(0)/h \leq x_k(0) + (1 - h)/h$. [DERIVED] □

A proposal, not a result. Because two populations cannot both exceed a threshold above one half while they sum to one, a first-passage record with $h > 1/2$ would give $\alpha = 0$ from conservation alone. This is a mathematical route to exclusive records. It is not a demonstrated detector model: it has not been implemented with local detectors, it presupposes the population martingale, and nothing in Section 3 or 5 supplies either.

Test. We used a discrete exchange kernel with $M = 2000$ units: $n' \sim \text{Multinomial}(M, x)$, so that $E[x' \mid x] = x$ exactly and vertices are absorbing [ASSUMED]. The kernel is global (each update uses all populations) and is not derived from Part I. With 20 000 trials per state, the Born test passed for all seven states, $n = 2, 3, 4$ (largest upper TV bound 0.019); no run was unresolved within 60 000 steps; and one-step drift measured at states visited along trajectories had $|z| > 3$ in 0.16–0.31% of components, consistent with the 0.27% expected by chance [TESTED]. Controls: a uniform start compared with the $(0.7, 0.2, 0.1)$ target was detected (TV lower bound 0.356).

Falsification arm. A bias b toward channel 0, $\tilde{x}_0 = x_0(1 + b)/(1 + bx_0)$, produces drift $b\,x_0(1 - x_0)/(1 + bx_0)$. For two channels, the diffusion approximation predicts the probability that channel 0 is recorded [7]:

$$u(p) = \frac{1 - e^{-2Mbp}}{1 - e^{-2Mb}}. \tag{3}$$

For $p = 0.2$ and $2Mb \in \{0, 0.1, 0.4, 1, 4\}$, the observed frequencies 0.199, 0.209, 0.235, 0.287, 0.565 lie within their 99% Wilson intervals of the predicted 0.200, 0.208, 0.233, 0.287, 0.561 (Fig. 3) [TESTED].

To first order, $u - p \approx Mb\,p(1 - p)$. Keeping $|u - p| \leq \varepsilon$ requires

$$|b| \leq \frac{\varepsilon}{M\,p(1 - p)}, \tag{4}$$

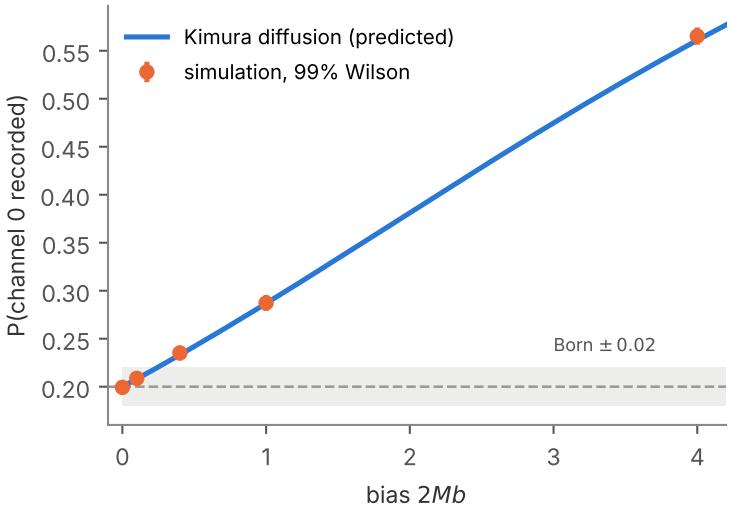


Figure 3: Biased exchange, two channels, $p = 0.2$, $M = 2000$. Points: simulated record frequency with 99% Wilson intervals (20 000 trials each). Line: Eq. (3), computed before the run. Grey band: Born value ± 0.02.

which is 6.3×10^{-5} at $M = 2000$, $\varepsilon = 0.02$, $p = 0.2$, and shrinks as $1/M$ [DERIVED]. The route is insensitive to the functional form of the exchange law but fragile in its *bias*: the drift must vanish to a precision set by the number of exchange events before absorption.

4.3 Comparison

The two routes fail in different ways. The threshold route needs an exact distributional property of each detector; the exchange route needs an exact symmetry of the exchange, to order $1/M$. Both are sufficient; we do not claim either is necessary for Born weights in general, only that within each route the stated condition is necessary (Yellott's theorem; Proposition 2). Collapse models obtain Born weights through the second structure [8, 9, 10]; the present analysis asks whether a classical substrate could supply it.

5 Question 3: does Part I supply these conditions?

5.1 The Part I representation is a permutation action

In the Part I helicity representation, a crossing of defects p and q exchanges their helicities, which acts on the eight helicity configurations of three defects by an 8×8 permutation matrix, independent of the crossing sign. The original source of this representation (`sfit_substrate_v8`) was not recoverable; it was reconstructed from the Part I→Part II integration code, which states the action explicitly.

Proposition 4 (Permutation no-go)**.** *Let $x(t+1) = P_{t+1}x(t)$ with P_{t+1} a random permutation matrix and $(\mathcal{F}_t)$ any filtration to which x is adapted. Then: (a) for every $x(t)$, with or without equal entries, the sorted vector of populations is invariant, so $\max_k x_k$ never changes and no population is concentrated; (b) if the entries of $x(t)$ are pairwise distinct, then $E[\Delta x \mid \mathcal{F}_t] = 0$ only if $P_{t+1} = I$ almost surely given $\mathcal{F}_t$. When entries coincide, a permutation that only exchanges equal entries*

leaves x unchanged, so (b) requires the distinctness hypothesis; (a) does not.

Proof. (a) A permutation rearranges entries. (b) $Q = E[P_{t+1} \mid \mathcal{F}_t]$ is doubly stochastic, and $E[\Delta x \mid \mathcal{F}_t] = (Q - I)x(t)$. Order the entries $x_{a_1} > x_{a_2} > \ldots$. Row a_1 of $Qx = x$ is an average of entries with weights summing to one that equals the largest entry, so $Q_{a_1 a_1} = 1$. Removing a_1 and repeating gives $Q = I$, and a convex combination of permutation matrices equals I only if all its weight is on I. [DERIVED] □

Conditioning more finely cannot help: the more the filtration determines the next crossing, the closer Q is to the actual permutation and the larger the drift. A physical helicity configuration is a basis vector, which a permutation maps to a basis vector, so the dynamics never produces a population vector with more than one nonzero entry. In this representation the eight components label configurations of the substrate, not measurement channels.

5.2 Measurement on the Part I crossing process

The Part I planar dynamics (three defects, speed 2, Poisson direction scattering at rate 2, harmonic trap 0.1) was integrated with $\Delta t = 0.005$ for $T = 20\,000$ in ten independent realisations, giving 8 403–9 026 order changes each [TESTED]. Of these, 0.03–0.10% were not single adjacent swaps; the permutation action is applied exactly as the net permutation, so none is lost. The four generators occurred with frequency 0.249–0.251 and the three defect pairs with 0.330–0.336. The signed one-crossing drift $E[\Delta x \mid x]$, evaluated at 200 fixed test states with 95% intervals across realisations, excluded zero at every test state; the largest component was -0.345 $[-0.349, -0.342]$. Conditioning on the defect ordering before the crossing increased the mean drift from 0.153 to 0.165–0.167, as Proposition 4 predicts. Along all ordered trajectories, the sorted population vector deviated from its initial value by exactly zero.

Positive control. The populations of the isotypic sectors of S_3 (trivial, rank 4; standard, rank 4), $\|\Pi_s \psi\|^2$, changed by at most 4.4×10^{-16} per crossing, as they must: the sector projectors commute with every element of the representation [DERIVED][TESTED].

5.3 Other representations

A two-dimensional representation $\rho(\sigma_1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\rho(\sigma_2) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, used in earlier exploratory work, satisfies the braid relation but is not orthogonal and was specified by hand. With populations renormalised after each crossing, the drift pulls x_1 toward 1/2, from $+0.233$ near $x_1 = 0$ to -0.230 near $x_1 = 1$ (Fig. 4) [TESTED]. For an orthogonal rotation by $\pm\varphi$ within a pair of channels, with both signs equally likely (as measured), $E[\Delta x_1 \mid x] = \sin^2\varphi\,(1 - 2x_1)$, which vanishes for all x only if $\sin\varphi = 0$ [DERIVED]. Rotations mix populations toward equality; they do not absorb.

5.4 Reading

Braiding, in every representation examined, either rearranges or mixes channel populations; it never concentrates them. In the sector basis it leaves populations exactly invariant. The natural reading is that braiding is neutral with respect to record selection, and that both the stochastic selection and the absorption into a record must come from the coupling between carrier and detectors, not from the braid action. This identifies where a derivation would have to look; it does not provide one.

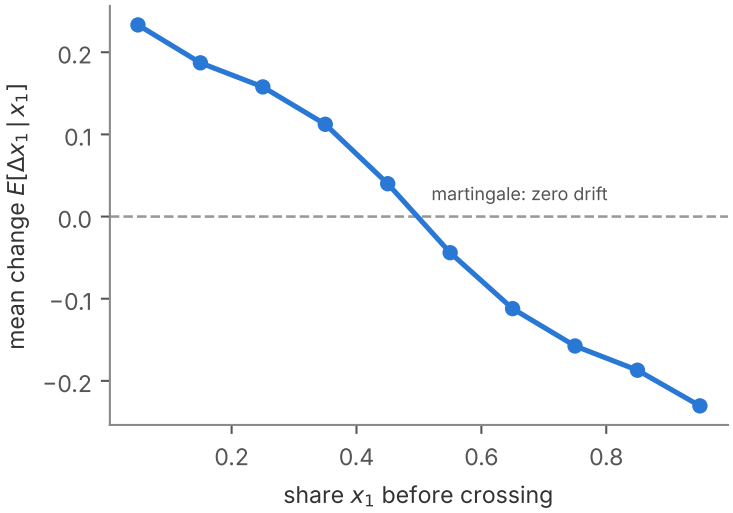


Figure 4: Mean one-crossing change of x_1 under the hand-specified two-dimensional B_3 representation, on the Part I crossing sequence. The drift pulls x_1 toward $1/2$ from both sides: the opposite of absorption. Intervals across ten realisations are smaller than the markers.

6 Limitations and open problems

The analysis does not derive quantum superposition or the Born rule from the Part I substrate. It establishes structural requirements for a record mechanism and identifies a limitation of the present helicity representation. Specifically:

1. The internal state ψ is an input. Part I does not produce states with more than one nonzero channel amplitude, and a sign-dependent unitary representation of B_3 would be a candidate ingredient, not by itself a derivation of physical superposition.
2. Why the physically relevant state space consists of measurement sectors rather than helicity configurations has not been shown.
3. The exchange kernel of Route 2 is global and postulated; a local, carrier-mediated version and the origin of its fairness (for example, from the symmetric reversals of Part I) remain to be derived, to the precision of Eq. (4).
4. Memoryless detector thresholds (Route 1) are assumed; their derivation from the substrate is open.
5. Strong exclusion was not achieved. The first-passage record of Proposition 3 would give $\alpha = 0$ by conservation, but it has not been combined with local detectors in a simulation.
6. Real amplitudes suffice for the single-system statistics studied here, but real-amplitude quantum theory is experimentally distinguishable from complex quantum theory in network scenarios [11].

7 Reproducibility

Three scripts produce every number in this paper. Each states its model, metrics, controls, predictions and decision rules in its header, fixed before the runs reported here, and prints its decisions as computed from the results. The scripts were not deposited before the runs, so the analysis is not preregistered in the strict sense; the predictions that failed are reported as failures. SHA-256 hashes of the versions run:

- `step0_braid_drift_v2.py` (Section 5):
  `8c935b8f8c99f18cfa8bb45e1ca6313c17b9674ce9654b545eb7d034d23dc07b`
- `part3_harness_v2.py` (Section 3, Route 1):
  `e5611f85c595798c3b9815c41d18610b1e36763f5e28bda54226304a91f9cbe4`
- `exchange_audit_v2.py` (Route 2):
  `dd06dbd4e7466a2f82ee3c5fe2fd88cb18c20b38d951d6ab00f58350992201ea`

The first two were run on two machines (Windows and Linux) with identical numerical output. The exchange audit was also run on both machines; the two copies had different file hashes and gave different numbers, with identical decisions on every test. The numbers reported are those of the version whose hash is listed. Code and results: `https://github.com/Dr-Loo/SFIT-Braid-Dynamics`.

References

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